EE2211 Introduction to Machine Learning

Lecture 2

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Office Hour (Hybrid): Monday 11:00am - 12:00nn (Week 2-4, Week 11-13)

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Course Contents

- Introduction and Preliminaries (Yueming)
  - Introduction
  - Data Engineering
  - Introduction to Probability and Statistics
- Fundamental Machine Learning Algorithms I (Shaojing)
  - Systems of linear equations
  - Least squares, Linear regression
  - Ridge regression, Polynomial regression
- Fundamental Machine Learning Algorithms II (Shaojing)
  - Over-fitting, bias/variance trade-off
  - Optimization, Gradient descent
  - Decision Trees, Random Forest
- Performance and More Algorithms (Yueming)
  - Performance Issues
  - K-means Clustering
  - Neural Networks

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Summary of Lec 1

Three Components in ML Definition

Task T, Performance P, Experience E

Two Types of Supervised Learning

Classification, Regression

Three Types of in ML

Supervised Learning
Unsupervised Learning
Reinforcement Learning

One Type of Unsupervised Learning

Clustering

Inductive and Deductive

Inductive: Probable
Deductive: Rule-based

Example of a Classifier Model

Nearest Neighbor Classifier

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Practice Question
(Type of Question to Expect in Exams)

Which of the following statement is true?

A. Nearest Neighbor Classifier is an example of unsupervised learning
B. Nearest Neighbor Classifier is an example of deductive learning
C. Nearest Neighbor Classifier is an example of feature extraction
D. None of the above is correct.

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Outline

Types of data

Data wrangling and cleaning

Data integrity and visualization

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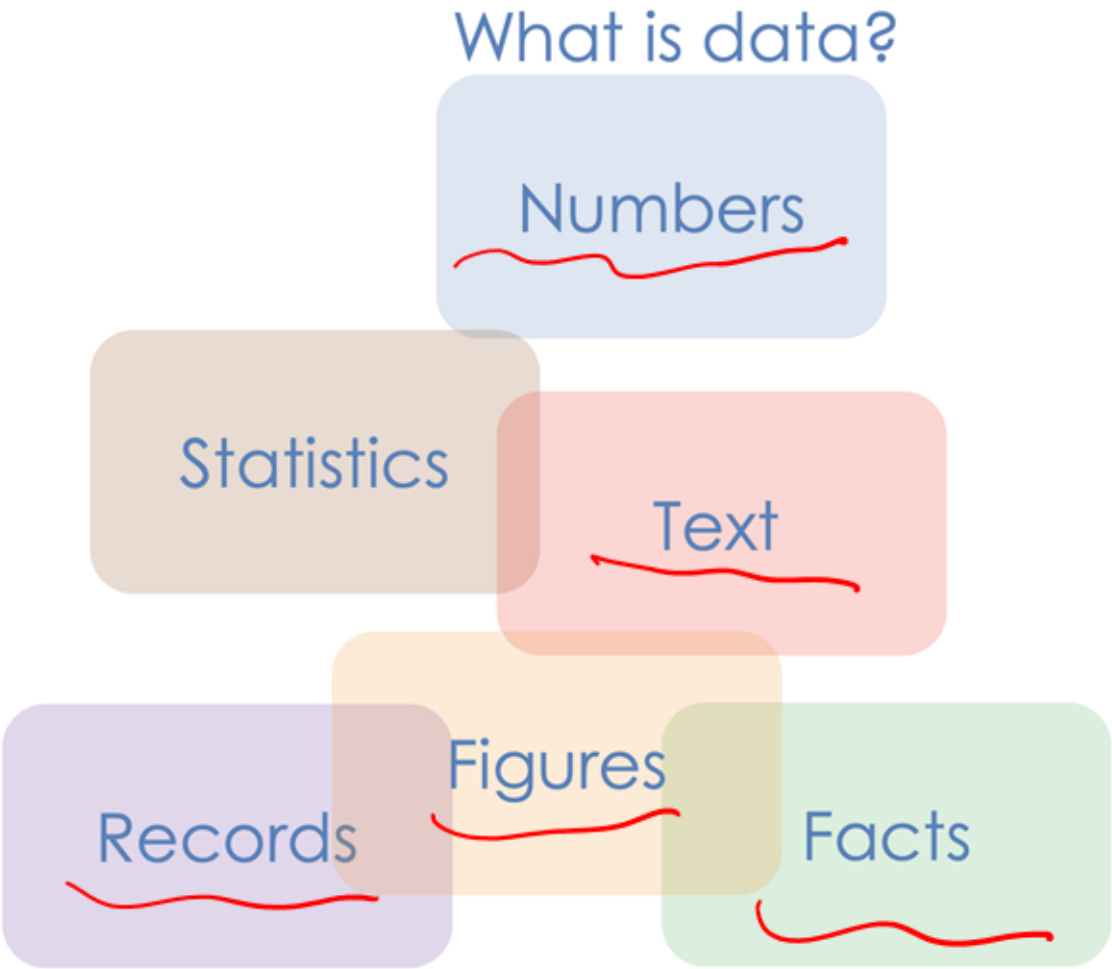
Types of Data



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Ways of Viewing Data

- Based on Levels/Scales of Measurement:
  - Nominal Data
  - Ordinal Data
  - Interval Data
  - Ratio Data
- Based on Numerical/Categorical
  - Numerical, also known as Quantitative
  - Categorical, also known as Qualitative
- Other aspects
  - Available or Missing Data

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Levels/Scales of Measurement

Ordinal

Named + Ordered

Nominal

Named

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Levels/Scales of Measurement

NOIR

Nominal

Named

Ordinal

Named + Ordered

Interval

Named + Ordered
+ Equal Interval

Ratio

Named + Ordered
+ Equal Interval +
Has "True" Zero

Highest

Lowest

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A Quick Recap: Mean, Median, Mode

- If we are given a sequence of numbers:

1, 3, 4, 6, 6, 7, 8

Mean: computing the average

(1+3+4+6+6+7+8)/7 = 5

Median: number in the middle (after sorting)

1, 3, 4, 6, 6, 7, 8

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A Quick Recap: Mean, Median, Mode

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Mean: computing the average

(1+3+4+6+6+7+8)/7 = 5

Median: number in the middle (after sorting)

1, 3, 4, 6, 6, 7, 8

*In case of even number of elements

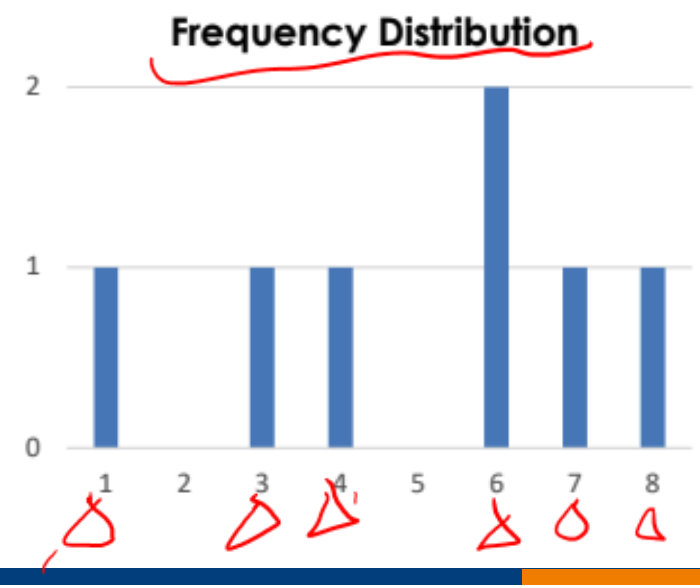
1, 3, 4, 6, 7, 8

(4+6)/2=5

Mode: number with highest frequency

1, 3, 4, 6, 6, 7, 8

6



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Nominal Data

- Lowest Level of Measurement
- Discrete Categories
- **NO** natural order
- Estimating a mean, median, or standard deviation, would be meaningless.
- Possible Measure: mode, frequency distribution

- Example:

Gender

1:man 2:woman

Occupation

Doctor Police Teacher

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Ordinal Data

NOIR

- **Ordered** Categories
- Relative Ranking
- Unknown "distance" between categories: orders matter but not the difference between values
- Possible Measure: mode, frequency distribution + median

- Example:
  - Evaluate the difficulty level of an exam
    - 1: Very Easy, 2: Easy, 3: About Right, 4: Difficulty, 5: Very Difficulty

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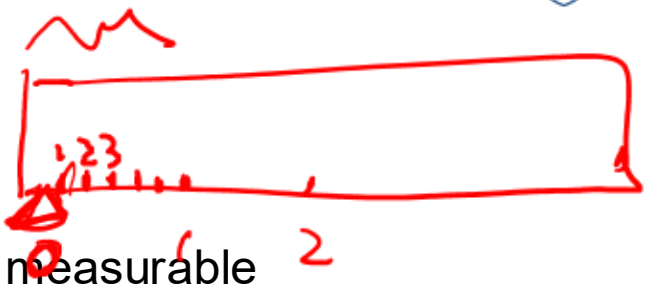
Interval Data



- Ordered Categories
- Well-defined "**unit**" measurement:
  - Distances between points on the scale are measurable
  - Can measure differences!
- **Equal Interval** (between two consecutive unit points)

- Example:
  - Temperature measured in Celsius
    - 10 degrees C, 20 degrees C, 30 degrees C
  - Year of someone's birth
    - 1990, 2000, 2010, 2020

3°C 4°C 5°C

1°C = 1°C

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Interval Data

- Ordered Categories
- Well-defined "**unit**" measurement:
  - Distances between points on the scale are measurable
  - Can measure differences!
- **Equal Interval** (between two consecutive unit points)
- **Zero is arbitrary** (not absolute), in many cases human-defined
  - If the variable equals zero, it does not mean there is none of that variable

- Example:
  - Temperature measured in Celsius
    - 10 degrees C, 20 degrees C, 30 degrees C
  - Year of someone's birth
    - 1990, 2000, 2010, 2020

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Interval Data

- Ordered Categories
- Well-defined "**unit**" measurement:
  - Distances between points on the scale are measurable
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- **Equal Interval** (between two consecutive unit points)
- **Zero is arbitrary** (not absolute), in many cases human-defined
  - If the variable equals zero, it does not mean there is none of that variable
- **Ratio is meaningless**
- Possible Measure: mode, frequency distribution + median + mean, standard deviation, addition/subtraction

$\frac{10+20}{2} = 15°C$

$\frac{\cancel{20°C}}{\cancel{10°C}} = 2$

$0°C \rightarrow$ defined by human

- Example:
  - Temperature measured in Celsius
    - 10 degrees C, 20 degrees C, 30 degrees C
  - Year of someone's birth
    - 1990, 2000, 2010, 2020

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NOZR

Ratio Data

- Most precise and highest level of measurement
- Ordered
- Equal Intervals
- **Natural Zeros:**
  - When the variable equals zero, it means there is none of that variable
  - Not arbitrary
- Possible Measure: mode, frequency distribution + median + mean, standard deviation, addition/subtraction + multiplication and division (ratio)

- Example:
  - Weights
    - 10 KG, 20 KG, 30 KG

0 kg

$\frac{20\text{ kg}}{10\text{ kg}} = 2$

  - Time
    - 10 Seconds, 1 Hour, 1 Day

0 second

$\frac{60\text{ sec}}{10\text{ sec}} = 6$

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NOIR

We can estimate	Nominal	Ordinal	Interval	Ratio
Frequency Distribution	Yes	Yes	Yes	Yes
Median	No	Yes	Yes	Yes
Add or subtract	No	No	Yes	Yes
Mean, standard deviation	No	No	Yes	Yes
Ratios	No	No	No	Yes

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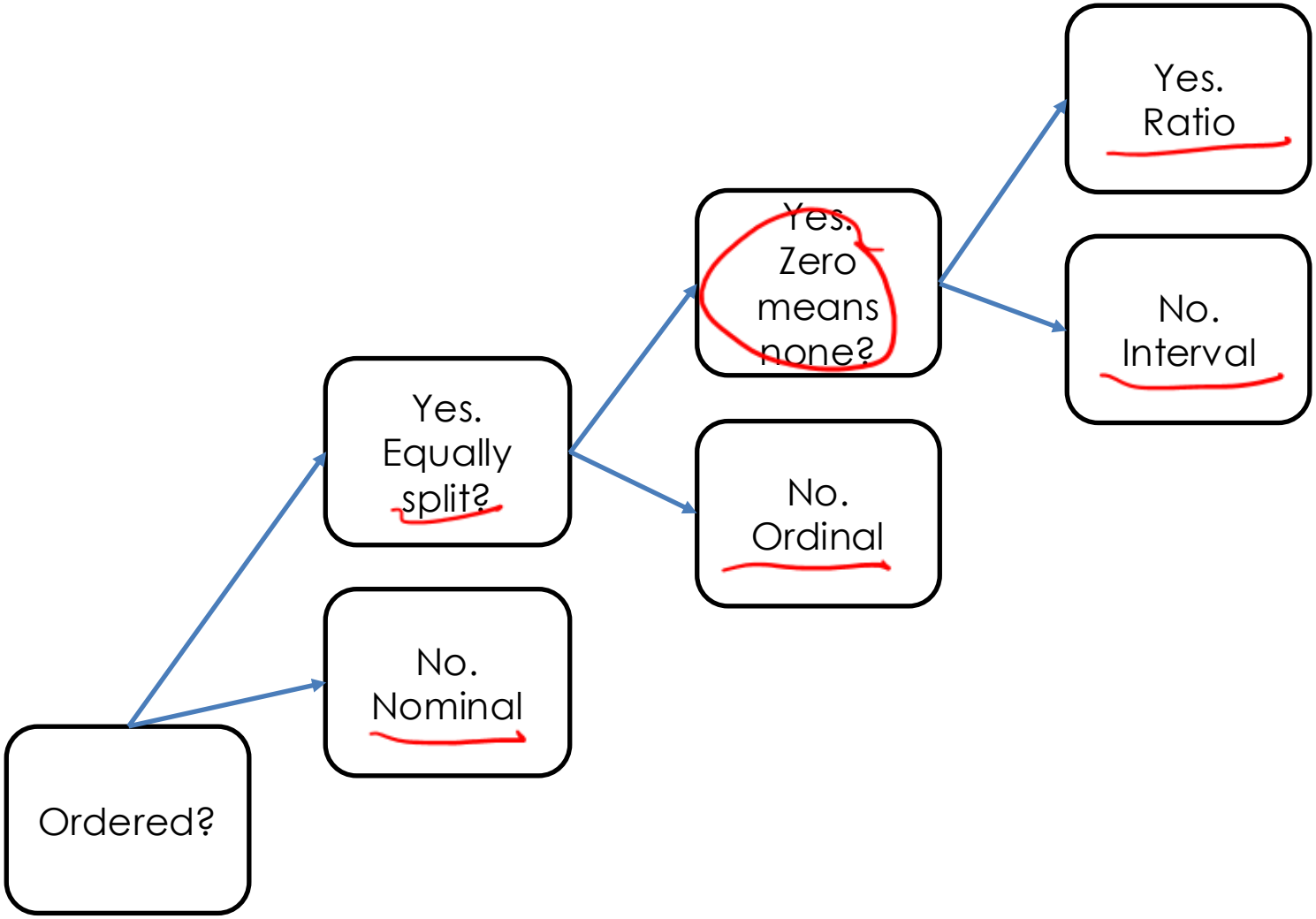
NOIR



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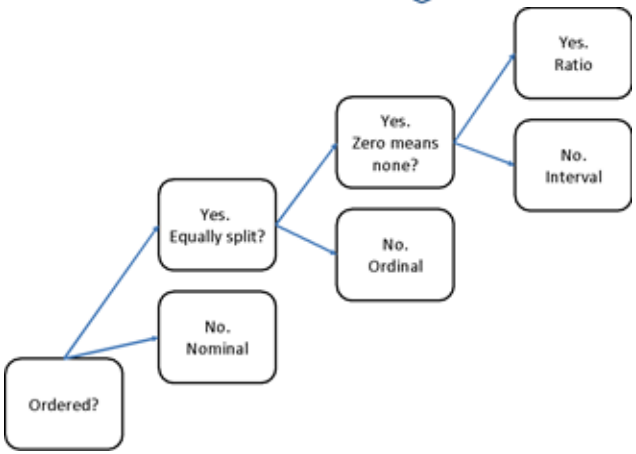
- Which level of measurement?

Nominal, Ordinal, Interval, Ratio

1. **Favorite Restaurant**
   - Mcdonald's, Burger King, Subway, KFC, … N
2. **Weight of luggage measured in KG** R
3. **SAT Scores: note that, SAT ranges is [400, 1600]**]
4. **Size of Packed Eggs in supermarkets** O
   - Small, Medium, Large, Extra Large, …



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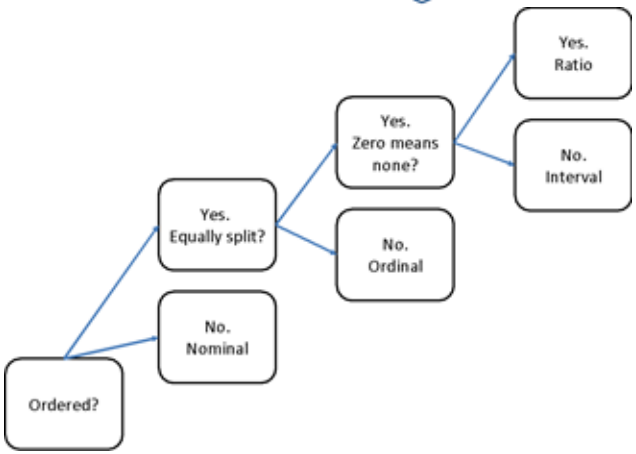
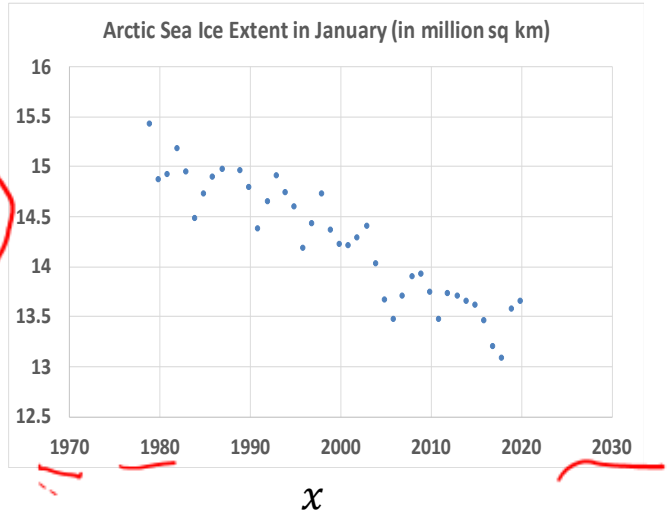
- Which level of measurement?

Nominal, Ordinal, Interval, Ratio



1. **Favorite Restaurant**
   - Mcdonald's, Burger King, Subway, KFC, …
2. **Weight of luggage measured in KG**
3. **SAT Scores: note that, SAT ranges is [400, 1600]**
4. **Size of Packed Eggs in supermarkets**
   - Small, Medium, Large, Extra Large, …
5. **Military rank**
   - General, Major, Captain, … O
6. **Number of people in a household** R
   - 1, 2, 3, 4, 5, …
7. **Credit Score in United States: the range is [300, 850]** I
8. 

R I

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Ways of Viewing Data

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- Based on Numerical/Categorical
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- Other aspects
  - Available or Missing Data

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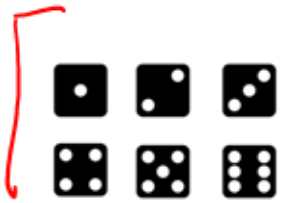
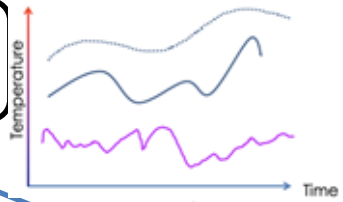
Numerical or Categorical

Variable

Categorical (qualitative)



Numerical (quantitative)



Nominal
(Unordered categories)

Ordinal
(Ordered categories)

Discrete
(Whole numerical values)
Example: outcome of tossing a die

Continuous
(Can take any value within a rage)
Example: temperature in day

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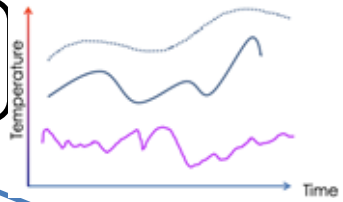
Numerical or Categorical

Variable

Categorical (qualitative)

Numerical (quantitative)



Nominal (Unordered categories)

Ordinal (Ordered categories)

Discrete (Whole numerical values) **Example:** outcome of tossing a die

Continuous (Can take any value within a rage) **Example:** temperature in day

OR

Numerical (quantitative)

Interval (may compute difference but no absolute zero)

Ratio (may compute difference, real zero exists)

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Ways of Viewing Data

- Based on Levels/Scales of Measurement:
  - Nominal Data
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1

- Based on Numerical/Categorical
  - Numerical, also known as Quantitative
  - Categorical, also known as Qualitative

2

- Other aspects
  - Available or Missing Data

3

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Missing Data

- Missing data: data that are missing and you do not know the mechanism.
  - You should use a single common code for all missing values (for example, “NA”), rather than leaving any entries blank.

NUS student	Age	Country of birth
Olivia Tan	20	Singapore
Hendra Setiawan	19	Indonesia
John Smith	19	NA

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Outline

Types of data

Data wrangling and cleaning

Data integrity and visualization

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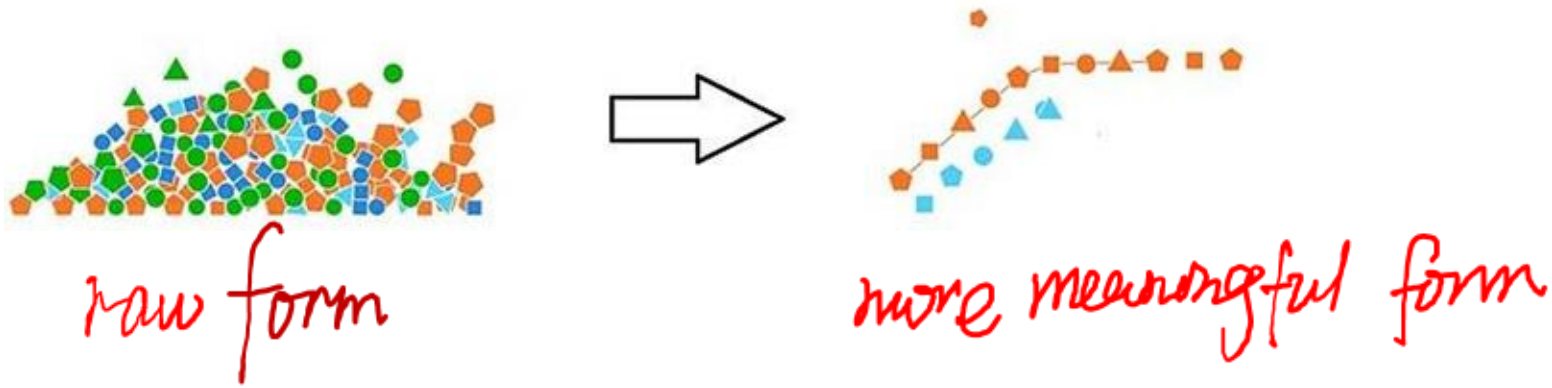
Data Wrangling

- Data wrangling
  - The process of transforming and mapping data from one "raw" data form into another format with the intent of making it more appropriate and valuable for a variety of downstream purposes such as analytics.
  - In short, transforms data to gain insight
  - It is a general process



Credit:https://en.wikipedia.org/wiki/Data_wrangling

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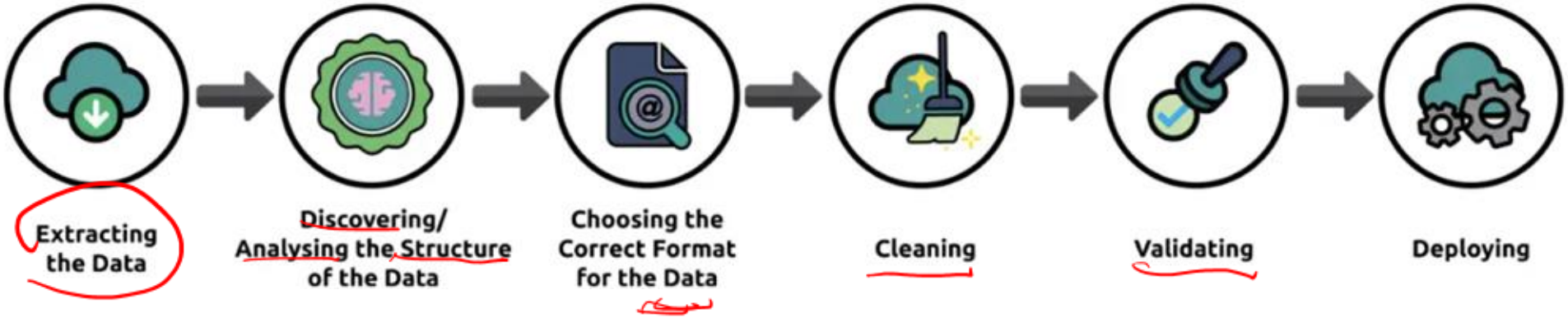
Data Wrangling



Credit:https://understandingdata.com/what-is-data-wrangling/

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Data Wrangling

Determine your goal

Make the most of dataset

Example: customer data stored in SQL; social network stored as graphs

Remove invalid data

Ensure data correctness

Use data (training/test)

happen before ML

Credit:https://understandingdata.com/what-is-data-wrangling/

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Data Wrangling

Determine your goal

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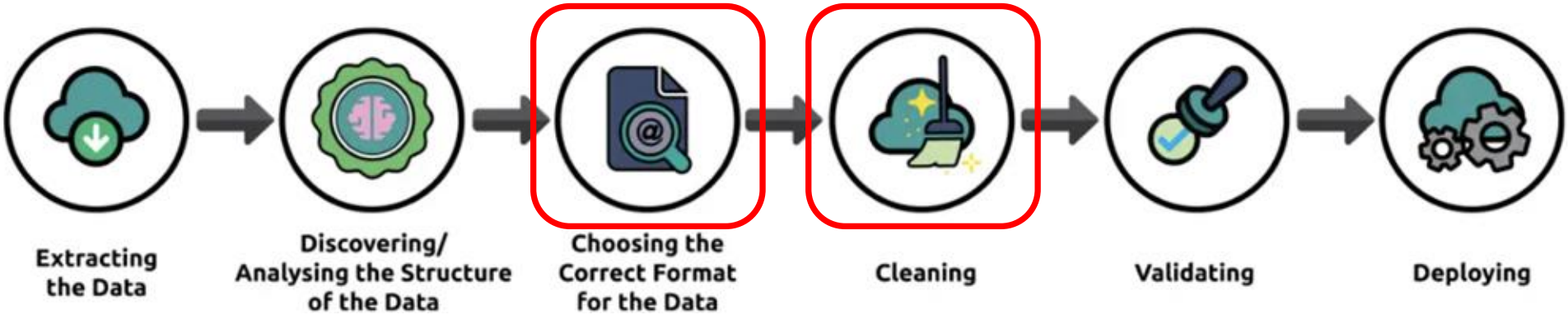

Women

Men

Unifying format to .png

Remove noisy samples

Checking that all images have labels.

Use data to train your face detector!

Collect Human Face Images for Face Detector

Credit:https://understandingdata.com/what-is-data-wrangling/

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Formatting Data

- **Binary Coding** to convert categories into binary form
  - One-hot encoding: unify several entities within one vector
    - Example: the color of a pixel can be red, yellow, or green
    - Very common in classification tasks!

$$\text{red} = [1, 0, 0]$$
$$\text{yellow} = [0, 1, 0]$$
$$\text{green} = [0, 0, 1]$$

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Formatting Data

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$$\text{red} = [1, 0, 0]$$
$$\text{yellow} = [0, 1, 0]$$
$$\text{green} = [0, 0, 1]$$

1x3

red y g

Y R G

[0 1 0]

[1 0 0]

[0 0 1]

- **Normalization**
  - Linear Scaling:
    scale each variable to [0 1]

$$x_i = \frac{x_i^{raw} - x^{min}}{x^{max} - x^{min}}, \quad i = 1, 2, ..., M$$

[1 3 5 9 11 1] 25

  - Z-score standardization:
    each independent dimension of data is normally distributed

$$x_i = \frac{x_i^{raw} - E[X]}{\sigma(X)}, \quad i = 1, 2, ..., M.$$

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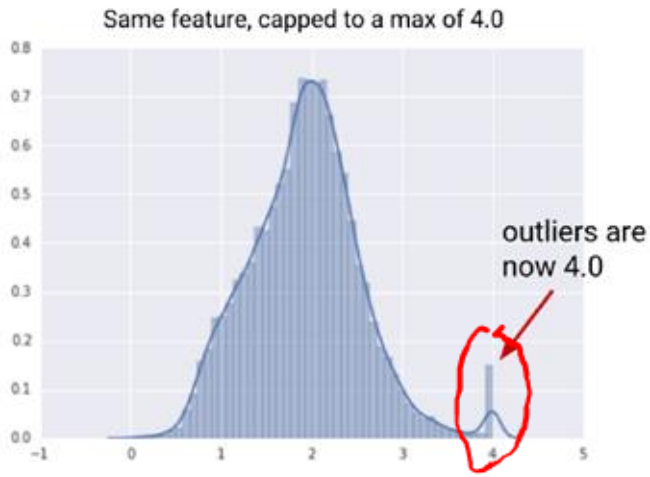
Data Cleaning

- The process of detecting and correcting (or removing) corrupt or inaccurate records from a record set, table, or database.
- Example:
  - Clipping outliers



  - Handling missing features

Students	Year of Birth	Gender	Height	GPA
Tan Ah Kow	1995	M	1.72	4.2
Ahmad Abdul	X NA	M	1.65	4.1
John Smith	1995	M	1.75	X
Chen Lulu	1995	F	X	4.0
Raj Kumar	1995	M	1.73	4.5
Li Xiuxiu	1994	F	1.70	3.8

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Data Cleaning: Handling missing features

1. Removing the examples with missing features from the dataset
   - Can be done if the dataset is big enough so we can sacrifice some training examples

2. Using a learning algorithm that can deal with missing feature values
   - Example: random forest

3. Using a data imputation technique

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Data Cleaning: Handling missing features: Imputation

- Method 1. Replace the missing value of a feature by an average value of this feature in the dataset:

$$\hat{x}^{(j)} \leftarrow \frac{1}{N} \sum_{i=1}^{N} x_i^{(j)}$$

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Data Cleaning: Handling missing features: Imputation

- Method 1. Replace the missing value of a feature by an average value of this feature in the dataset:

$$\hat{x}^{(j)} \leftarrow \frac{1}{N}\sum_{i=1}^{N} x_i^{(j)}$$

- Method 2. Highlight the missing value ✓ commonly used
  - Replace the missing value with a value outside the normal range of values.
  - For example, if the normal range is [0, 1], then you can set the missing value to 2 or −1.
  - Enforce the learning algorithm to learn what is best to do when the feature has a value significantly different from regular values.

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Outline

Types of data

Data wrangling and cleaning

Data integrity and visualization

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Data Integrity

- Data integrity is the maintenance and the assurance of data accuracy and consistency;
  - A critical aspect to the design, implementation, and usage of any system that stores, processes, or retrieves data.
  - Very broad concept!
- Example:
  - In a dataset, numeric columns/cells should not accept alphabetic data.
  - A binary entry should only allow binary inputs

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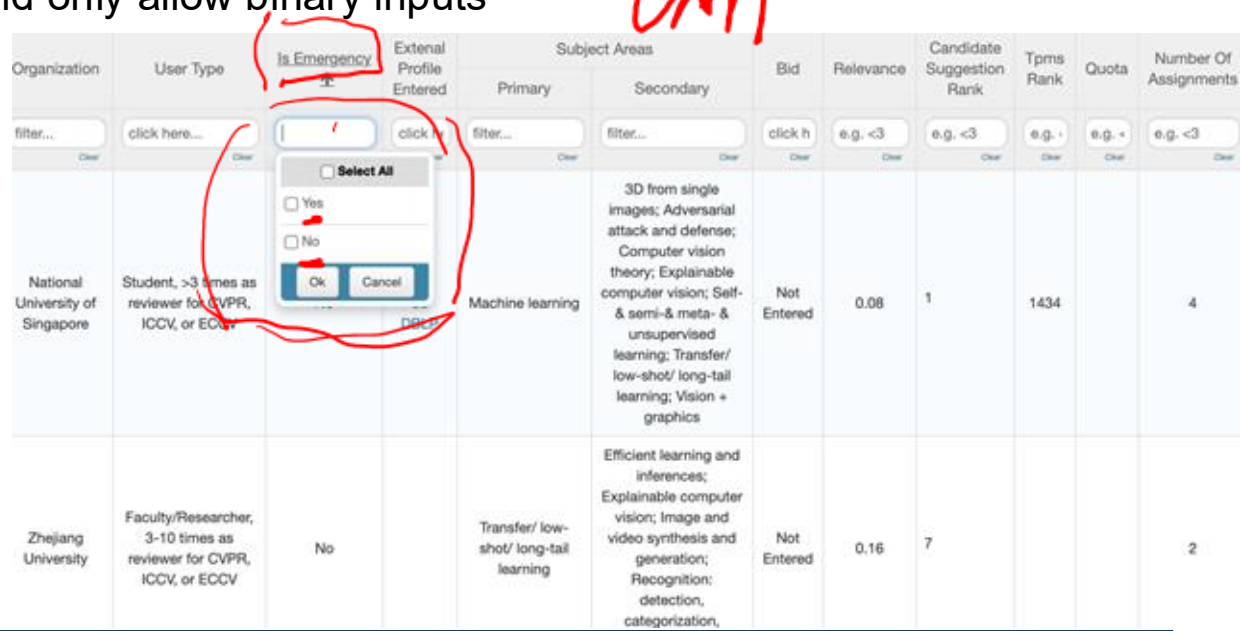
CMT



Organization	User Type	Is Emergency	Extenal Profile Entered	Subject Areas: Primary	Subject Areas: Secondary	Bid	Relevance	Candidate Suggestion Rank	Tpms Rank	Quota	Number Of Assignments
filter...	click here...		click h	filter...	filter...	click h	e.g. <3	e.g. <3	e.g.	e.g.	e.g. <3
National University of Singapore	Student, >3 times as reviewer for CVPR, ICCV, or ECCV			Machine learning	3D from single images; Adversarial attack and defense; Computer vision theory; Explainable computer vision; Self- & semi-& meta- & unsupervised learning; Transfer/ low-shot/ long-tail learning; Vision + graphics	Not Entered	0.08	1	1434		4
Zhejiang University	Faculty/Researcher, 3-10 times as reviewer for CVPR, ICCV, or ECCV	No		Transfer/ low-shot/ long-tail learning	Efficient learning and inferences; Explainable computer vision; Image and video synthesis and generation; Recognition: detection, categorization,	Not Entered	0.16	7			2

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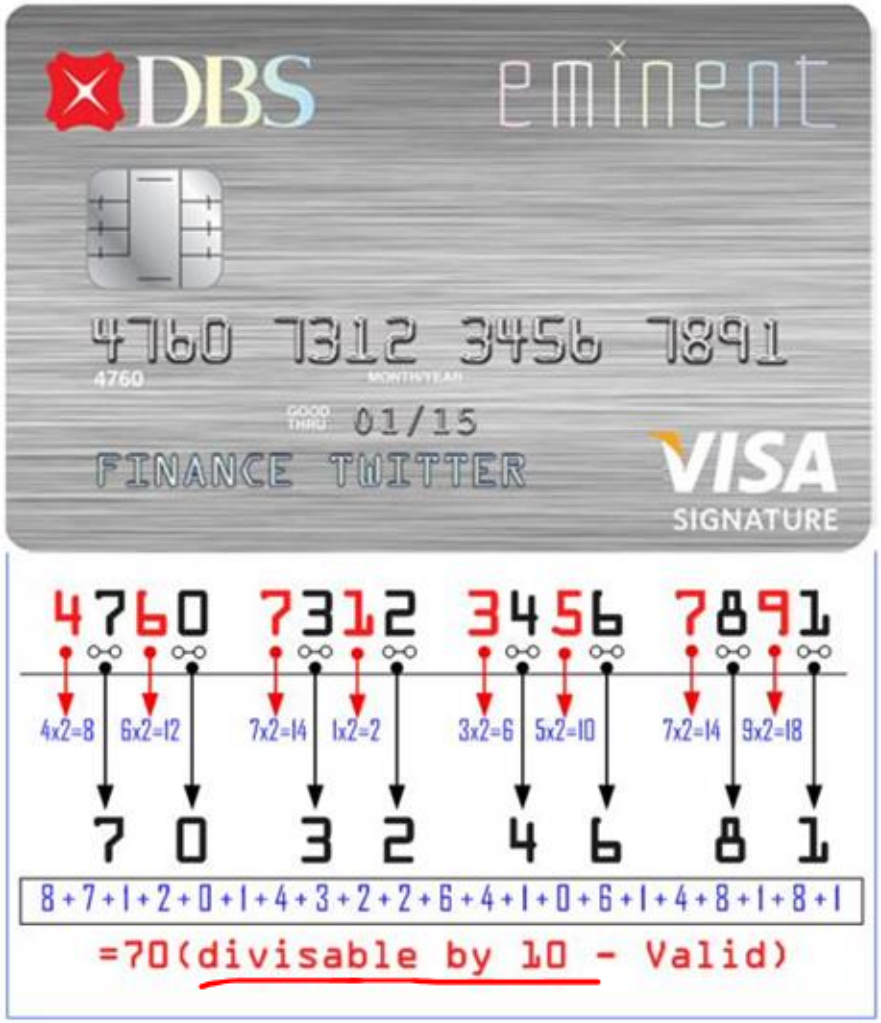
Data Integrity



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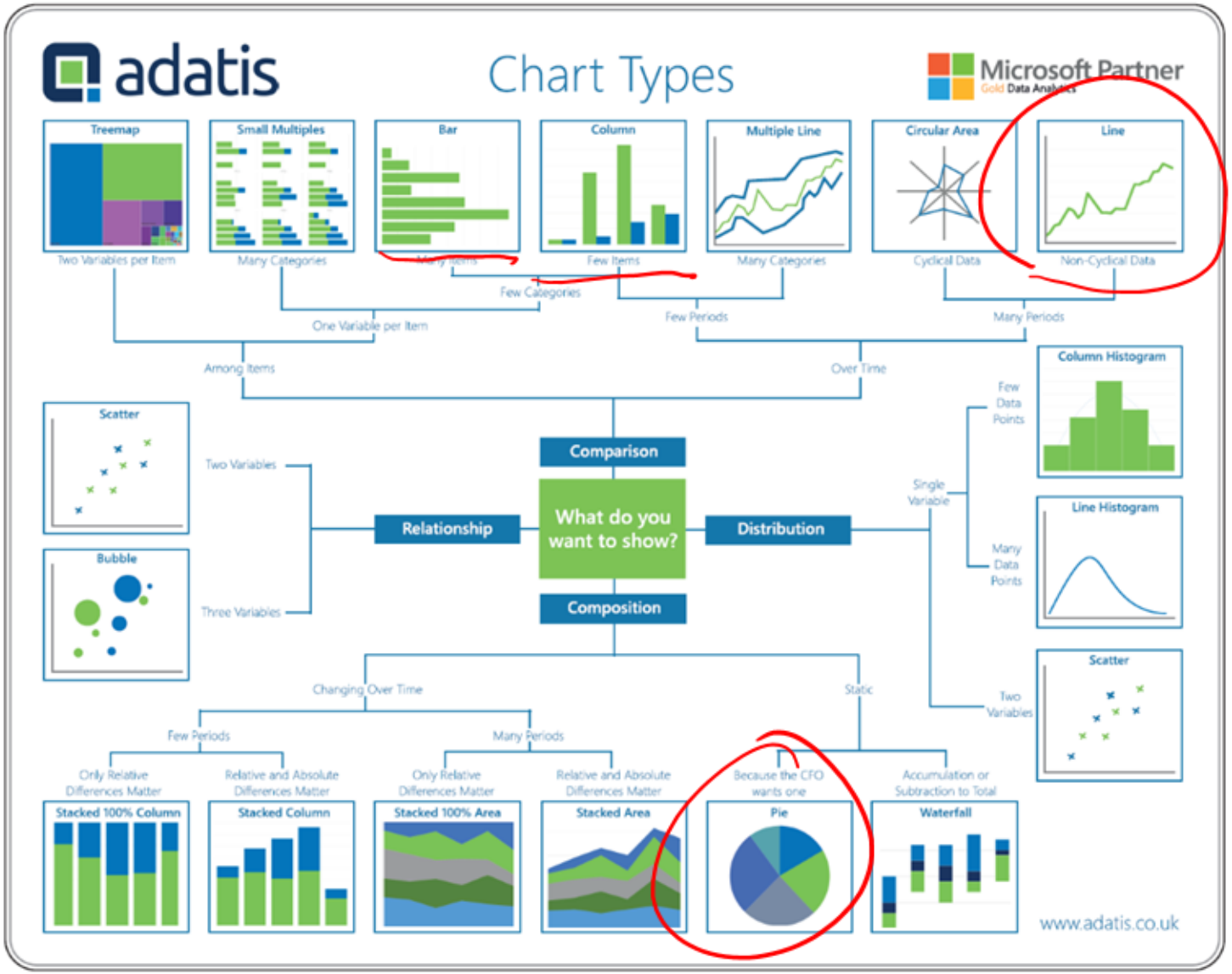
Data Visualization



Graphical Representation of data!

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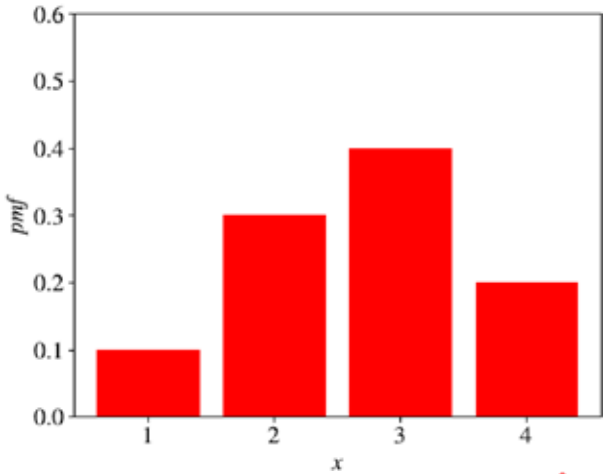
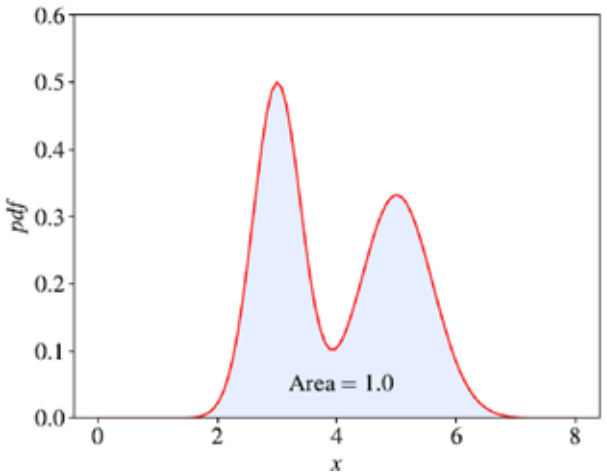
Visualization: Distribution



Probability Mass Function

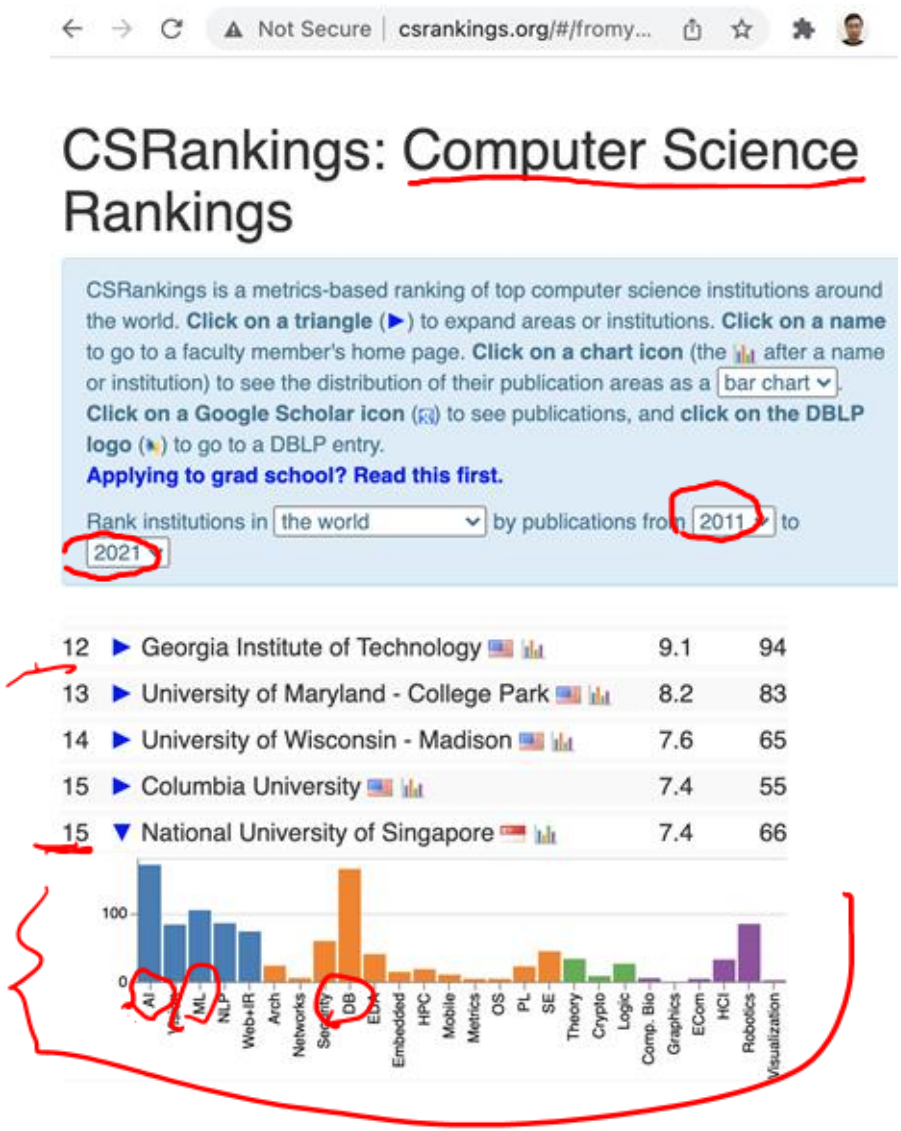
PMF



Probability Density Function

PDF

Visualization: Bars



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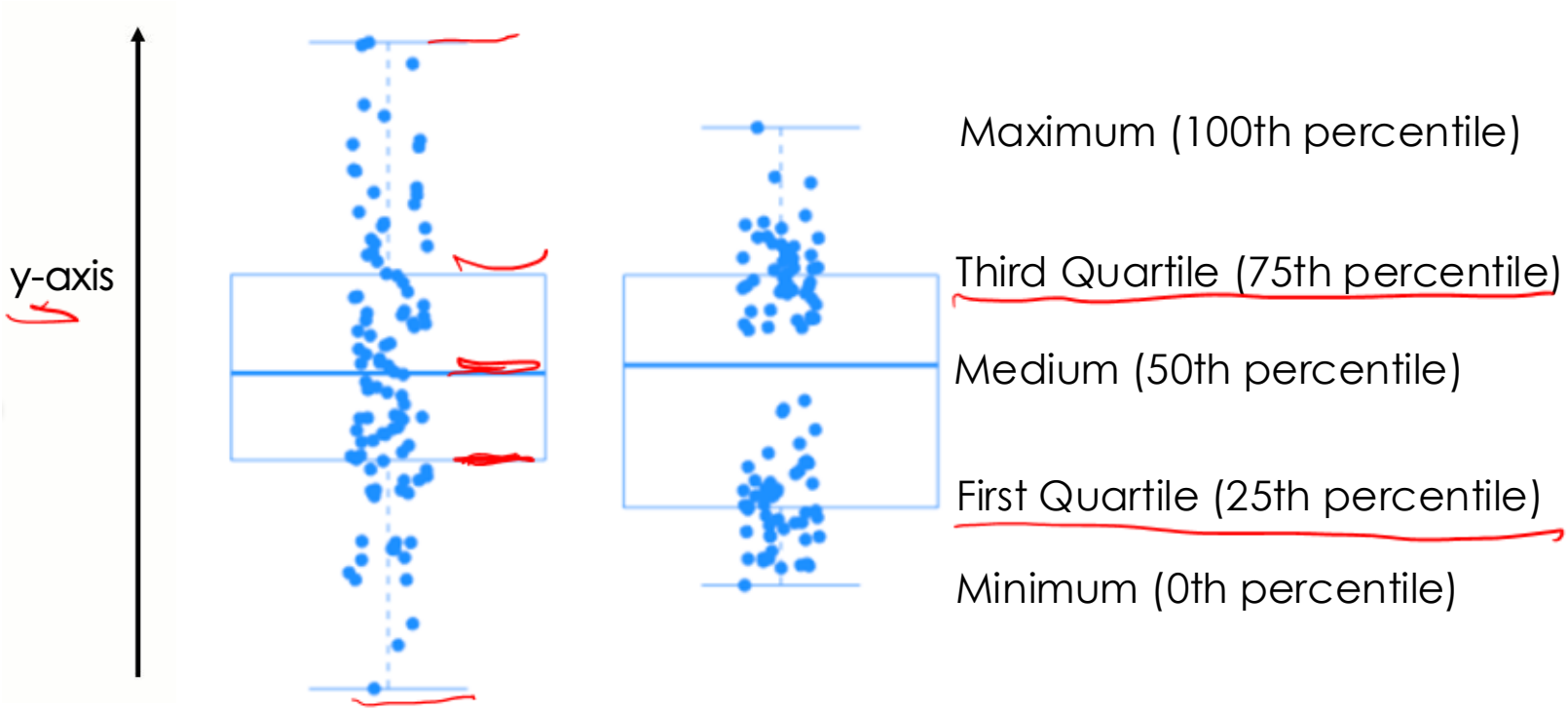
Visualization: Boxplots



- The first quartile (Q_1) is defined as the middle number between the smallest number (i.e., Minimum) and the median of the data set.

- The third quartile (Q3) is the middle number between the median and the highest value (i.e., Maximum) of the data set.

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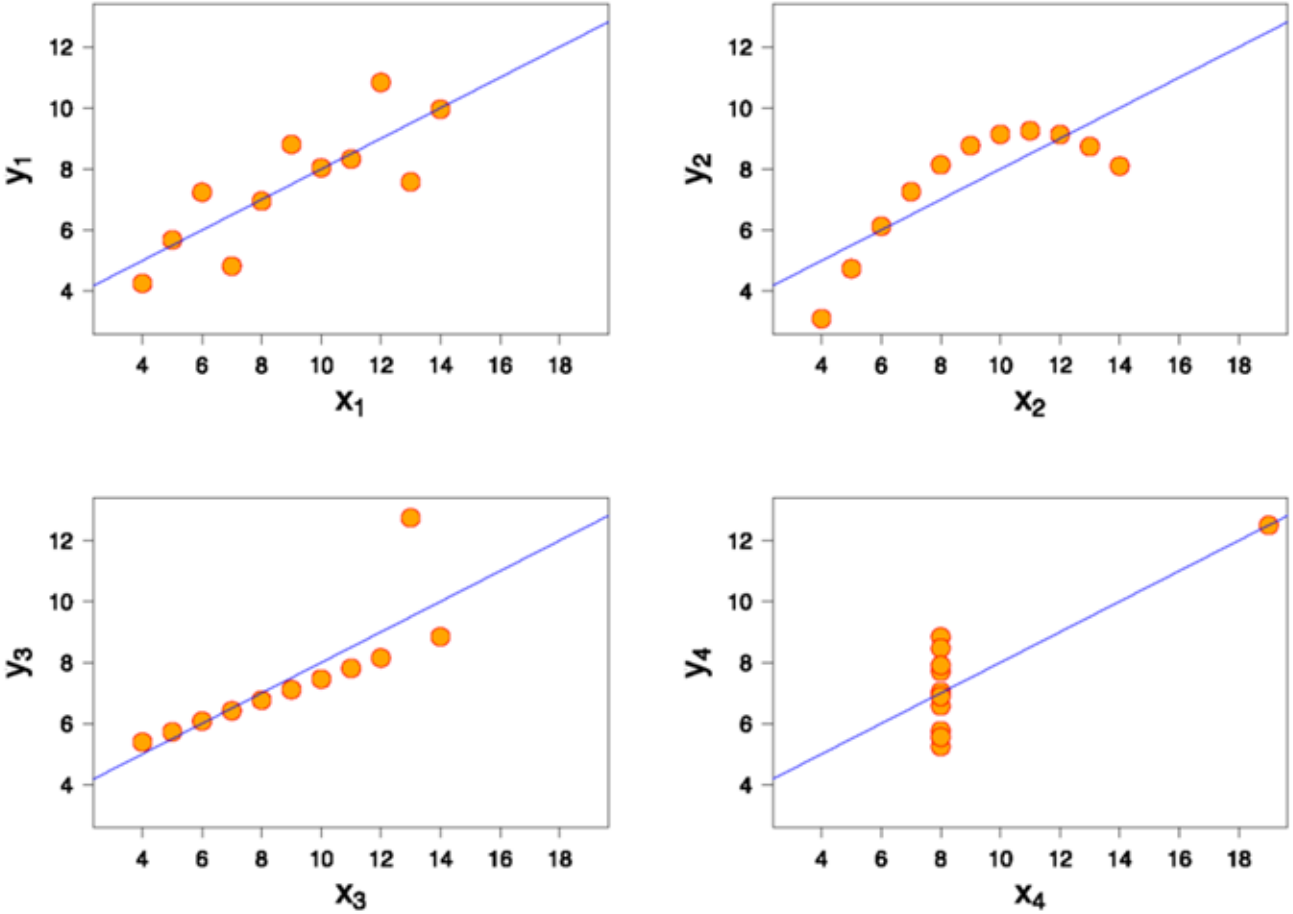
Why Visualization is Necessary



Four datasets with **identical** means, variances and regression lines.

Hence, we need visualization to show their difference!

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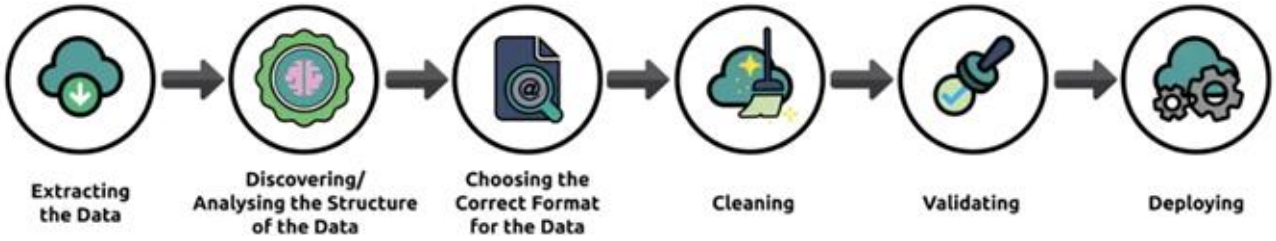
Summary

- Types of data
  - NOIR
- Data wrangling and cleaning



- Data integrity and visualization
  - Integrity: Design
  - Visualization: Graphical Representation

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Practice Question
(Type of Question to Expect in Exams)

N	O	N
Color	**Size**	**Shape**
Blue	Large	Ring
Red	Large	Triangle
Orange	Large	Diamond
Green	Small	Circle
Yellow	Small	Arrow
Blue	Large	Rectangle
Red	Large	Circle
Green	Small	Diamond

What are the NOIR data types of color, size, and shape in the table?

What about their label, yes/no?

N

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